

Date: 8 April

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Lecture

Circular Convolution by DFT

Suppose that we have two finite duration sequences of length N , $x_1(n)$ and $x_2(n)$. Their respective N point DFT are

$$X_1(k) = \sum_{n=0}^{N-1} x_1(n) e^{-j \frac{2\pi n k}{N}} \rightarrow (P)$$

$k = 0, 1, 2, \dots, N-1$

$$X_2(k) = \sum_{n=0}^{N-1} x_2(n) e^{-j \frac{2\pi n k}{N}} \rightarrow (P)$$

$k = 0, 1, 2, \dots, N-1$

If we multiply the two DFTs together the result is a DFT say $X_3(k)$ of a sequence $x_3(n)$ of length N .

Let us determine the relationship between $x_3(n)$ and the sequence $x_1(n)$ & $x_2(n)$. We have

$$X_3(k) = X_1(k) X_2(k) \quad k = 0, 1, \dots, N-1$$

The IDFT of $X_3(k)$ is:

$$x_3(m) = \frac{1}{N} \sum_{k=0}^{N-1} X_3(k) e^{j \frac{2\pi k m}{N}}$$



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$$= \frac{1}{N} \sum_{k=0}^{N-1} x_1(k) x_2(k) e^{j 2\pi k m} \rightarrow (iii)$$

Suppose that we substitute for $x_1(k)$ & $x_2(k)$ in eq iii

using the DFT's given in eq (i) & eq (ii)
Thus we obtain

$$x_3(m) = \frac{1}{N} \sum_{k=0}^{N-1} \left[\sum_{n=0}^{N-1} x_1(n) e^{-j 2\pi k n} \right] \cdot \left[\sum_{l=0}^{N-1} x_2(l) e^{-j 2\pi k l} \right] e^{j 2\pi k m}$$

$$= \frac{1}{N} \sum_{n=0}^{N-1} x_1(n) \sum_{l=0}^{N-1} x_2(l) \left[\sum_{k=0}^{N-1} e^{j 2\pi k (m-n-l)} \right] \rightarrow (iv)$$

The inner sum in the brackets in (iv) has the form

$$\sum_{k=0}^{N-1} a^k \quad \left\{ \begin{array}{ll} N & a = 1 \\ \frac{1-a^N}{1-a} & a \neq 1 \end{array} \right.$$

where a is defined as

$$a = e^{j 2\pi (m-n-l)}$$



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where $l = m - n$

$$a = \frac{e^{j2\pi(m-n-l)/N}}{N} = 1$$

when $l \neq m - n$

$$1 - \frac{e^{j2\pi(m-n-l)/N}}{N} \times N = 0$$

$$1 - \frac{e^{j2\pi(m-n-l)/N}}{N}$$

Hence

$$\sum_{k=0}^{N-1} a^k = \begin{cases} N & l = m - n \\ 0 & l \neq m - n \end{cases}$$

So

$$x_3(m) = \frac{1}{N} \sum_{n=0}^{N-1} x_1(n) \sum_{l=0}^{N-1} x_2(l)$$

→ remove this summation
 b/c only 1 value
 $l = m - n$

for $l = m - n$

$$x_3(n) = \sum_{n=0}^{N-1} x_1(n) x_2(m-n)$$

Form of
 circular
 convolution



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$$x_3(n) = \sum_{k=0}^{N-1} x_1(k) x_2(n-k)$$

Example 7.2.1

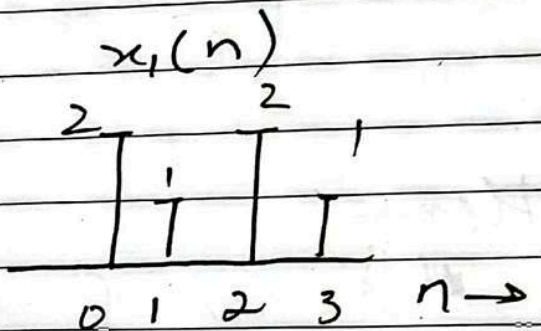
Perform the circular convolution of the following two sequences

$$x_1(n) = \{2, 1, 2, 1\}$$

$$x_2(n) = \{1, 2, 3, 4\}$$

$$m = 0$$

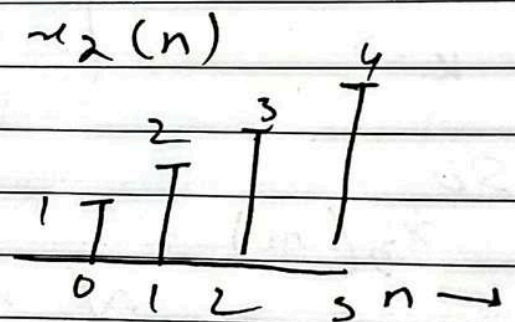
$$x_3(0)$$



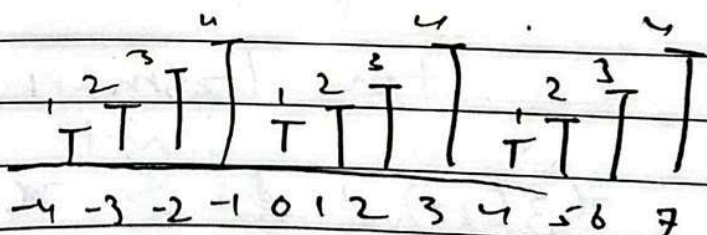
$$x_3(m) = \sum_{n=0}^{N-1} x_1(n) x_2(m-n)$$

$$= x_1(0)x_2(0) + x_1(1)x_2(-1) + x_1(2)x_2(-2) + x_1(3)x_2(-3)$$

$$= 2 \cdot 1$$

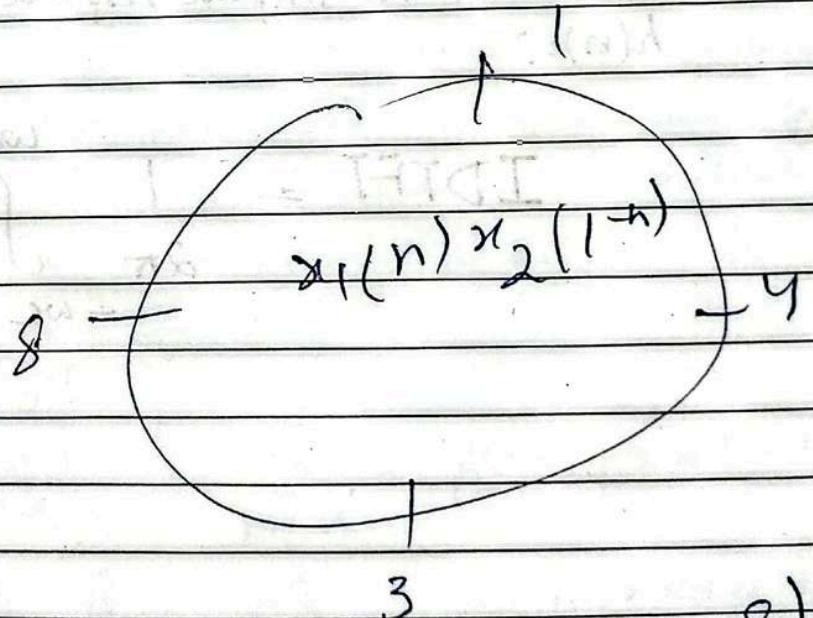
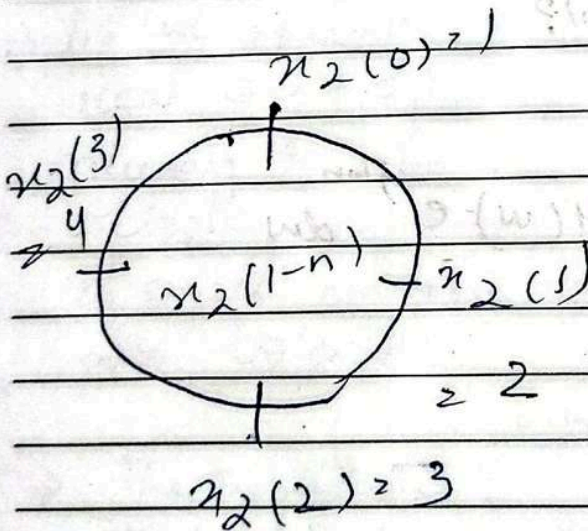
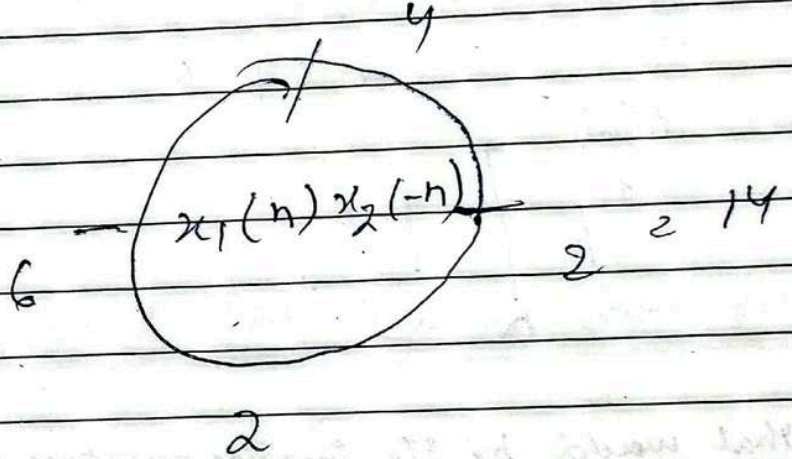
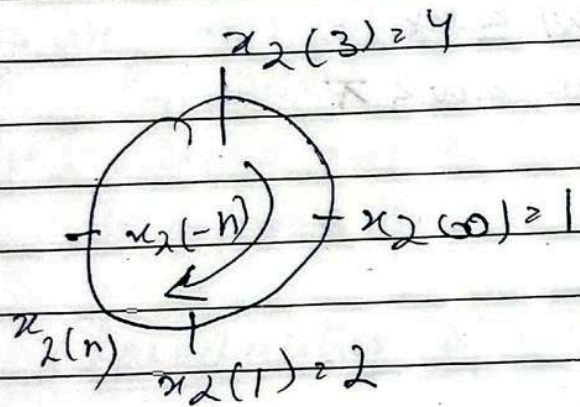
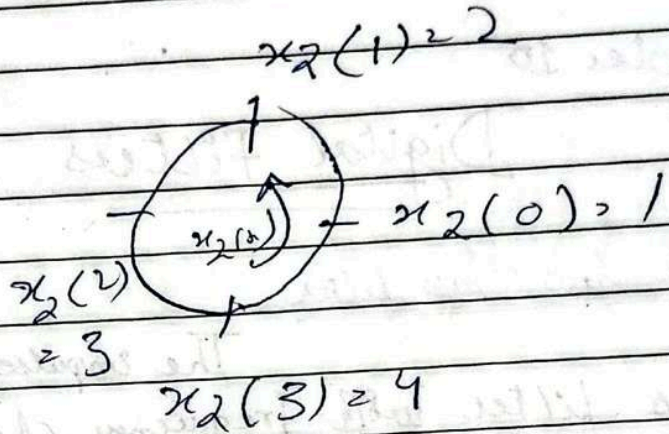
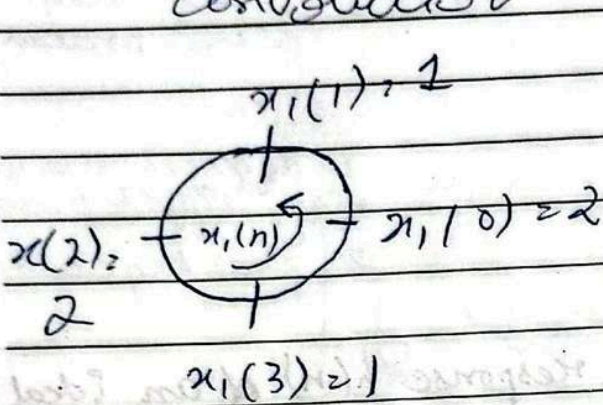


$$= \{14, 16, 17, 16\}$$



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Another method to perform circular convolution



$$\sum (1, 3, 4, 8)$$

$$x_3(n) = [14, 16, 19, 16]$$

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Lecture

Chapter 10

Digital Filters

Analysis

→ already design system product

Design

→ different groups apply different con on some key condi proved.

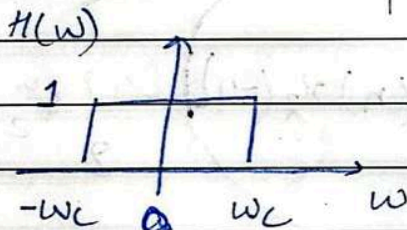
* Ideal Low Pass filter

The impulse response $h(n)$ of an ideal

Low pass filter with frequency characteristics is

$$\text{frequency character } H(\omega) = \begin{cases} 1 & |\omega| \leq \omega_c \\ 0 & \omega_c < \omega \leq \pi \end{cases}$$

of ideal low pass filter



What would be its impulse response $h(n)$?

$h(n)$?

$$\text{IDTFT} = \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} H(\omega) e^{j\omega n} d\omega$$

$$= \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} 1 e^{j\omega n} d\omega$$

$$= \frac{1}{2\pi} \left[\frac{e^{j\omega n}}{jn} \right]_{-\omega_c}^{\omega_c}$$

$$= \frac{1}{2\pi} \frac{e^{j\omega_c n} - e^{-j\omega_c n}}{jn}$$

$$= \frac{1}{2\pi} \frac{\sin\left(\frac{n\omega_c}{\pi}\right)}{\frac{n\omega_c}{\pi}} \quad n \neq 0$$



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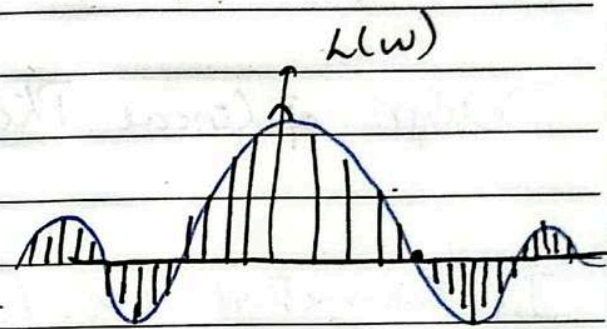
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For $n=0$ ω_c

$$h(0) = \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} e^{j\omega_0} d\omega = \frac{\omega_c}{\pi}$$

$$h(n) = \begin{cases} \frac{\omega_c}{\pi} & n=0 \\ \frac{1}{2\pi} \frac{\sin(\omega_c n)}{n} & n \neq 0 \end{cases}$$



Unit sample response of an IDFT

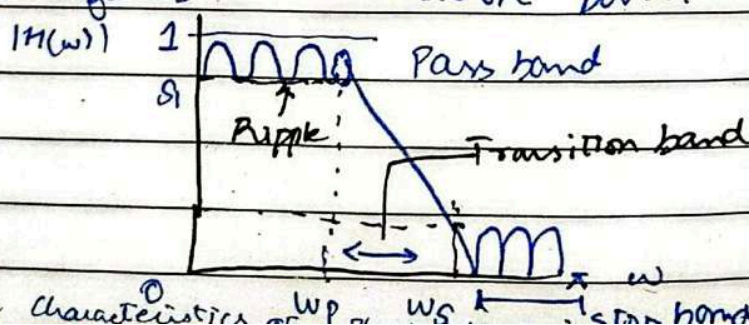
Q. Why ideal low pass filter is not physically realizable?

A. Its impulse response has infinite points we need infinite data to implement it

Characteristics of Practical Frequency selective frequency

⇒ As we observed that ideal filters are unrealizable for real time signal processing applications they are not absolutely necessary in most practical applications. A small amount of ripple is usually tolerable in passband. Similarly a small non zero value or small amount of ripple is also tolerable in stopband.

⇒ The transition of the frequency response from pass band to stopband defines the transition band.



ω_p = edge of passband

ω_s = edge of stopband



Magnitude characteristic of low pass filter

DIGITAL FILTERS

Date: 15th April 2025

* Ideal low pass filter:

→ The impulse response $h(n)$ of an ideal low pass filter with frequency characteristics is

$$H(\omega) = \begin{cases} 1 & |\omega| \leq \omega_c \\ 0 & \omega_c < \omega \leq \pi \end{cases}$$

$$\Rightarrow h(n) = \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} 1 e^{j\omega n} d\omega$$

$$\text{IDTFT} = \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} H(\omega) e^{j\omega n} d\omega$$

$$= \frac{1}{2\pi} \left[\frac{e^{j\omega n}}{jn} \right]_{-\omega_c}^{\omega_c}$$

$$= \frac{1}{2\pi} \left[\frac{e^{j\omega_c n}}{jn} + \frac{e^{-j\omega_c n}}{jn} \right] = \frac{1}{2\pi} \left[\frac{e^{j\omega_c n} + e^{-j\omega_c n}}{jn} \right]$$

$$h(n) = \frac{1}{2\pi} \cdot 2 \frac{\sin(n\omega_c)}{n} = \frac{\sin(n\omega_c)}{\pi n}$$

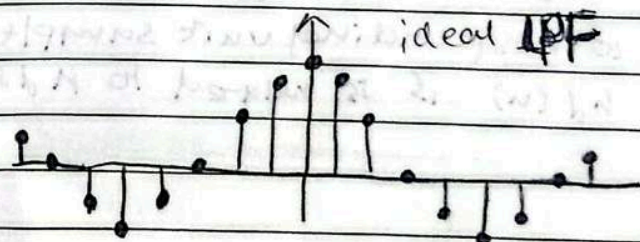
for $n=0$,

$$h(0) = \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} e^{j\omega \cdot 0} d\omega = \frac{\omega_c}{\pi}$$

→ Ideal low pass filter

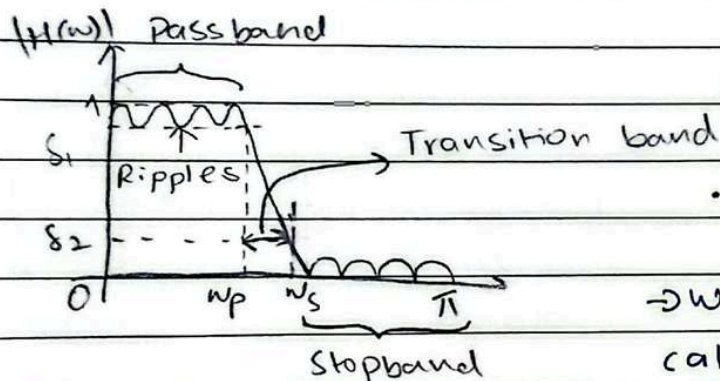
is not physically realizable as its impulse response has infinite points.

$$h(n) = \begin{cases} \omega_c / \pi & n=0 \\ \frac{\sin(n\omega_c)}{\pi n} & n \neq 0 \end{cases}$$



→ Characteristics of Practical Frequency-selective circuit filters:

- As we observed ^{the} ideal filters are unrealizable for real-time signal processing applications. They are not absolutely necessary in most practical applications. A small amount of ripple is usually tolerable in passband. Similarly, a small non-zero value or small amount of ripple is also tolerable in stopband.
- The transition of the frequency response from passband to stopband defines the transition band.



• ω_p = Edge of the passband.

• ω_s = Edge of the stopband.

→ width of the passband is called bandwidth of the filter.

→ Magnitude characteristics of physically realizable filters

→ Design of Linear-Phase FIR filters using windows:

- In this method we begin with the desired frequency response specification $H_d(w)$ & determine the corresponding unit sample response $h_d(n)$. (Indeed $h_d(n)$ is related to $H_d(w)$ by the Fourier Transform.

$$H_d(\omega) = \sum_{n=-\infty}^{\infty} h_d(n) e^{-j\omega n}$$

where

$$h_d(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} H_d(\omega) e^{j\omega n} d\omega \quad \text{--- (1)}$$

→ Thus, given $H_d(\omega)$, we can determine $h_d(n)$ by evaluating the integral in eq (1)

→ In general, $h_d(n)$ is infinite in duration & must be truncated at some point, say at $n = M-1$, to

- Truncation of $h_d(n)$ to a length $M-1$ is equivalent to multiplying $h_d(n)$ by a "rectangular window" defined as

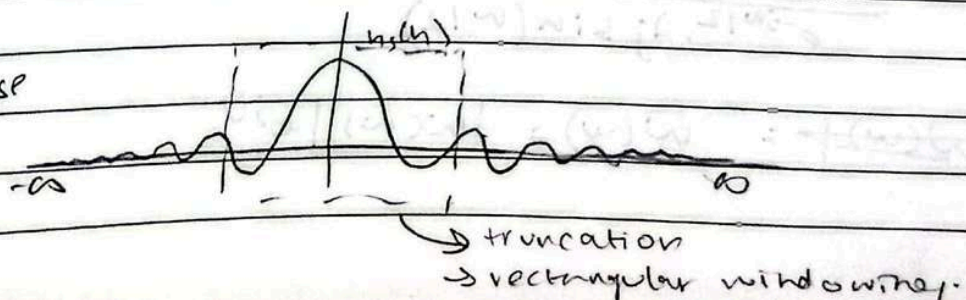
$$w(n) = \begin{cases} 1 & n = 0, 1, \dots, M-1 \\ 0 & \text{otherwise} \end{cases}$$

Thus, the unit sample response of the FIR filter becomes.

$$h(n) = h_d(n)w(n)$$

$$h(n) = \begin{cases} h_d(n) & n = 0, 1, \dots, M-1 \\ 0 & \text{otherwise} \end{cases}$$

IIR
infinite impulse response



Chapter 8 eth edition (design of digital filters)

Date: 22 April s plane $\rightarrow$ digital signal

Assuming $\rightarrow$ use have $\rightarrow$ analog to digital

lecture

Characteristics of commonly used Analog filters

Step for construction of digital signal

we need to convert it to z plane

1/R digital filters can easily be obtained by beginning with an analog filter and then using a mapping to transform the s plane into z plane. Thus, the design of digital filter is reduced to designing an appropriate analog filter and then performing the conversion from $H(s)$ to $H(z)$ plane.

Butterworth filter

Low pass Butterworth filters are [all pole filters] characterized by the following magnitude squared frequency response

$\downarrow$ frequency response of Butterworth filter where N is the

no zero have only poles	$ H(\Omega) ^2 = \frac{1}{1 + (\frac{\Omega}{\Omega_c})^{2N}}$	order of the filter Ω_c & the -3dB cut off frequency
-------------------------	--	---

Since $H(s)H(-s)$ evaluated at $s = j\Omega$ is simply equal to $|H(\Omega)|^2$ at frequencies that

$|H(\Omega)|^2 = H(s)H(-s) = \frac{1}{1 + (\frac{-s^2}{\Omega_c^2})^N} = \frac{1}{1 + (\frac{-s^2}{\Omega_c^2})^N}$

they both have out 1 equal

Poles are imp for stability of system

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$$= \frac{1}{1} \quad \frac{H(z)}{H(z)}$$

Step 1: transforming $\frac{D}{N_c}$ to $\frac{-s^2}{\Omega_c^2}$
From 8.3.48 we find that

$$-\frac{s^2}{\Omega_c^2} = (-1)^{\frac{1}{N}} = e^{j(2k+1)\frac{\pi}{N}}$$

and hence

$$k=0, 1, 2, \dots, N-1$$

$$Skz \Omega_c e^{j\frac{\pi}{2}} e^{j(2k+1)\frac{\pi}{2N}}$$

Step for transforming $\frac{D}{N_c}$ to $\frac{-s^2}{\Omega_c^2}$
 $|H(\Omega)|^2 = H(s)H(-s) = \frac{1}{\Omega_c^2}$

$$1 + \left(\frac{\Omega}{\Omega_c}\right)^{2N}$$

$$D = \sqrt{\Omega_c^2}$$

$$\frac{1}{\Omega_c^2 - s^2} = \frac{1}{1 + \left(\frac{\sqrt{\Omega_c^2 - s^2}}{\Omega_c}\right)^{2N}} = \frac{1}{1 + \left(\frac{\sqrt{-s^2}}{\Omega_c}\right)^{2N}}$$

$$S = j\omega \quad s^2 = -\omega^2 \quad \Omega_c^2 - s^2 = -\Omega_c^2$$

$$\boxed{s^2 - \Omega_c^2}$$

$$\boxed{-s^2 - \Omega_c^2}$$



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$$\frac{1}{1 + \left(\frac{-s^2}{\sqrt{2}c^2}\right)^N}$$

$$H\left(\frac{s}{\sqrt{2}c}\right)^N = \frac{-s^2}{\sqrt{2}c^2}$$

For finding poles put denominator equal to zero

$$|H(\sqrt{2})|^N = H(s)H(-s) = \frac{1}{1 + \left(\frac{-s^2}{\sqrt{2}c^2}\right)^N}$$

Find equation for poles

$$1 + \left(\frac{-s^2}{\sqrt{2}c^2}\right)^N = -1$$

$$\left(\frac{-s^2}{\sqrt{2}c^2}\right)^N = (-1)^{1/N}$$

$$\sqrt{2}c^2 = e^{j\pi} = e^{j(2k+1)\pi}$$

$$-s^2 = e^{j\pi} = e^{j(2k+1)\pi}$$

for $k=0,1,\dots,N-1$

Since adding

2πk doesn't

change any

$$-1 = e^{j\pi}$$

we know



Since $e^{j\pi} = \cos\pi + j\sin\pi = -1$
 Since $e^{j\pi} = e^{j(\pi + 2\pi k)}$
 For $k=0,1,\dots,N-1$
 Taking common

If you need to design digital filter

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$$-s^2 = \sqrt{2} e^{j(2k+1)\frac{\pi}{N}}$$

$$S^2 = \sqrt{2} e^{j(2k+1)\frac{\pi}{N}}$$

Taking sq root $\rightarrow$ of whole term

$$S = \sqrt{2} e^{j\frac{\pi}{N}} \sqrt{e^{j(2k+1)\frac{\pi}{N}}}$$

$$S_k = \sqrt{2} e^{j\frac{\pi}{N}} e^{j(2k+1)\frac{\pi}{2N}}$$

$\hookrightarrow k \rightarrow$ bcz at every value of k there will be an S

$$k \geq 0 \Rightarrow S_0$$

$\hookrightarrow$ They are the poles of frequency response

$\hookrightarrow$ Equation of poles for butter worth filter

$$|H(\omega)|^2 = |H(s)H(-s)| = \frac{1}{1 + \left(\frac{-s^2}{\sqrt{2}}\right)^N} \rightarrow (8.3.47)$$

$$1 + \left(\frac{-s^2}{\sqrt{2}}\right)^N$$

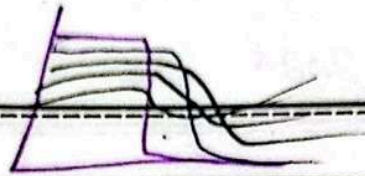
The order of the filter can be obtained from

eq (8.3.47) We want " N " to be have attenuation S_2 at a specified frequency ω_2

Thus $\alpha \propto \sqrt{2} \omega_2$

if increase order it start reading ideal curve

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$$= \frac{1}{1 + \left(\frac{\Omega_s}{\Omega_c}\right)^{2N}} \Rightarrow s^2 \Rightarrow \frac{1}{s^2} = \frac{1}{T\left(\frac{\Omega_c}{\Omega_s}\right)^{2N}}$$

stop band
frequency
edge
frequency

$$\left(\frac{\Omega_s}{\Omega_c}\right)^{2N} = \frac{1}{s^2} - 1 \rightarrow \text{Take natural log}$$

$$2N \log\left(\frac{\Omega_s}{\Omega_c}\right) = \log\left(\frac{1}{s^2} - 1\right)$$

if we want order of

System we need $N = \frac{\log\left(\frac{1}{s^2} - 1\right)}{2 \log\left(\frac{\Omega_s}{\Omega_c}\right)}$ at $\Omega = 0$

→ cut off frequency

→ stop band

→ attenuation

then we find poles

of system

Example 8.3.6

$$2 \log\left(\frac{\Omega_s}{\Omega_c}\right)$$

at $\Omega = \Omega_c$

$$H(\Omega_c)^2 = \frac{1}{2}$$

at $\Omega = \Omega_c$

$$\left(\frac{\Omega_c}{\Omega}\right)^{2N} \Rightarrow |H(\Omega)|^2 \rightarrow \text{approaching to zero}$$

→ Numerator

→ SK frequency

Determine the order and poles of a lowpass butterworth filter that has 3dB bandwidth of 500 Hz and an attenuation of 40dB at 100 Hz

→ cut off frequency
→ attenuation



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Lecture

Butterworth filter

Magnitude of frequency response $= |H(\Omega)|^2 = \frac{1}{1 + \left(\frac{\Omega}{\Omega_c}\right)^{2N}} = \frac{1}{1 + \left(\frac{-s^2}{\Omega_c^2}\right)^N}$

$\Omega_c = -3dB / \text{cut off frequency}$

poles of frequency response $s_k = \Omega_c e^{j\frac{\pi}{2N}} e^{j(2k+1)\frac{\pi}{2N}}$, for $k=0, 1, 2, \dots, N-1$

$N = \log \left[\frac{1}{\epsilon^2} - 1 \right]$

greater response become closer to $2 \log \left(\frac{\Omega_s}{\Omega_c} \right)$

ideal response

$|H(\Omega)|^2 = 1$

$1 + \left(\frac{\Omega}{\Omega_c}\right)^{2N}$

$\Rightarrow$ At $\Omega=0 \rightarrow \text{dc frequency}$
 $|H(0)|^2 = \frac{1}{1+0} = 1$

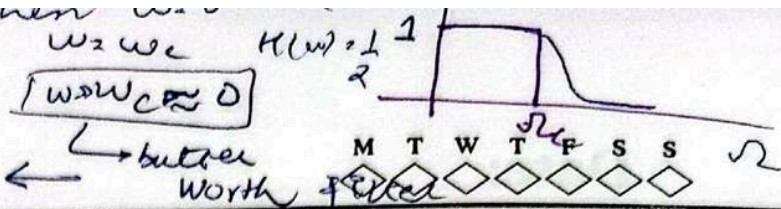
$\Rightarrow$ At $\Omega = \Omega_c \rightarrow \text{cut off frequency}$
 $|H(\Omega_c)|^2 = \frac{1}{1+1} = \frac{1}{2}$

$1 + \left(\frac{\Omega_c}{\Omega_c}\right)^{2N} = \frac{1}{2}$
 $1 + 1^{2N} = \frac{1}{2}$



attenuate
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behaving
like
stopband



At $\Omega \gg \Omega_c$ let $\Omega = 1000 \Omega_c = 1$

$$|H(\Omega)|^2 = \frac{1}{1 + \left(\frac{\Omega}{\Omega_c}\right)^{2N}} = \frac{1}{1 + \left(\frac{1000}{1}\right)^{2N}} \approx \left(\frac{1}{1000}\right)^{2N} = \left(\frac{\Omega_c}{\Omega}\right)^{2N}$$

① $\rightarrow$ we can neglect it

this term becomes nearly zero

Hence $|H(\Omega)|^2$ approaches zero as $\Omega \gg \Omega_c$ so for higher frequencies than Ω_c $|H(\Omega)|^2$ behaves like stopband.

Example 8.1.36

Determine the order and the poles of a Lowpass Butterworth filter that has a -3dB bandwidth of 500 Hz and an attenuation of 40 dB at 1000 Hz

$$\Omega_c = \text{cut off frequency} = 500 \text{ Hz} = 500(2\pi) = 1000$$

$$\Omega_s = 1000 \text{ Hz} = 2000$$

$$N = ?$$

$$N = \frac{\log\left(\frac{1}{\delta^2} - 1\right)}{2 \log\left(\frac{\Omega_s}{\Omega_c}\right)}$$

$$20 \log\left(\frac{1}{\delta}\right) = \text{dB}$$

$$20 \log\left(\frac{1}{\delta}\right) = 40$$

$$\log\left(\frac{1}{\delta}\right) = 2$$

$$|S = 0, 0|$$



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$$N = \log \left(\frac{1}{(0.01)^2} - 1 \right) \approx 6.64 \approx 7$$

$$2 \log \left(\frac{2000}{1000} \right)$$

$$S_k = \sqrt{c} e^{j\frac{\pi}{2}} e^{j(2k+1)\frac{\pi}{2N}} \quad k = 0, 1, \dots, N-1$$

$$S_k = 1000 e^{j\frac{\pi}{2}} e^{j(2k+1)\frac{\pi}{2N}} \quad k = 0, 1, \dots, 6$$

↑
 ache poles

LecturePoles of Butterworth (BW) of filters

$$S_k = \Omega_c e^{j\frac{\pi}{2}} e^{j\frac{(2k+1)\pi}{2N}} \quad (\text{equation for poles of BW filter})$$

$$|S_k| = \Omega_c e^{j\frac{\pi}{2}} e^{j\frac{(2k+1)\pi}{2N}} \quad \underbrace{\hspace{10em}}_1$$

$$k = 0, 1, 2, \dots, N-1$$

$$|S_k| = \Omega_c$$

Poles of BW filter lie on a circle of radius Ω_c

Let $N=4$ means 4 poles

$\Omega_c = 1$ for normalised case

$$k = 0, 1, 2, 3$$

$\Rightarrow k=0$
 For first pole $S_0 = \Omega_c e^{j\frac{\pi}{2}} e^{j\frac{(1)\pi}{2 \times 4}}$
 $= \Omega_c e^{j\frac{\pi}{2}} e^{j\frac{\pi}{8}} = 1 e^{j(\frac{\pi}{2} + \frac{\pi}{8})}$
 $\underbrace{\hspace{10em}}_{\theta}$

$$k=0 \quad \theta_0 = \frac{\pi}{2} + \frac{\pi}{8} = \frac{5\pi}{8}$$

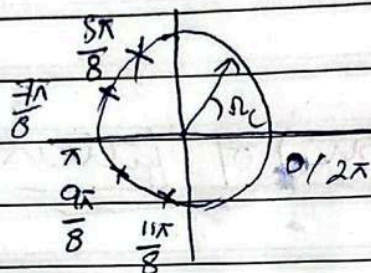
$$k=1 \quad \theta_1 = \frac{7\pi}{8}$$



$$\theta_2 = \frac{9\pi}{8}$$

$$\theta_3 = \frac{11\pi}{8}$$

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that's how poles of butterworth filter lie on circle of radius R_c

Chebyshev filters

Type 1

$$|H(\Omega)|^2 = \frac{1}{1 + \epsilon^2 T_N^2\left(\frac{\Omega}{\Omega_p}\right)}$$

(Frequency Response)

$\frac{\Omega}{\Omega_p}$ → given frequency
 Ω_p → passband edge frequency

ϵ = Parameter related to the ripple in the passband

$T_N(x)$ = N^{th} order of polynomial define as

$$T_N(x) = \begin{cases} \cos(N \cos^{-1} x) & |x| \leq 1 \\ \cosh(N \cosh^{-1} x) & |x| > 1 \end{cases}$$

if given frequency and passband edge frequency meet condition of this then they will be the x of T_N

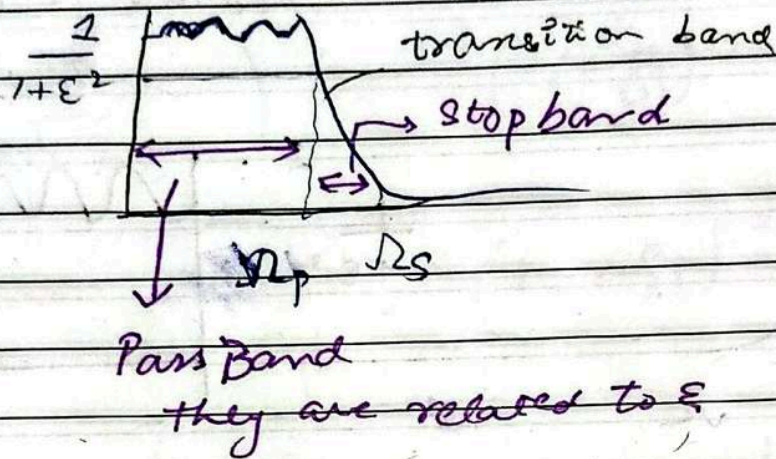
Recursion equation

$$T_{N+1}(x) = 2x T_N(x) - T_{N-1}(x)$$



where $T_0(x) = 1$ $T_1(x) = x$
 $N=0 \rightarrow \cos(0) = 1$

$N = 1, 2, \dots$



using Recursion equation

$$T_{N+1}(x) = 2xT_N(x) - T_{N-1}(x)$$

put $N=1$

$$T_2(x) = 2xT_1(x) - T_0(x)$$

$$= 2x(x) - 1$$

$$T_2(x) = 2x^2 - 1$$

Put $N=2$

$$T_3(x) = 2xT_2(x) - T_1(x)$$

$$= 2x(2x^2 - 1) - x$$

$$= 4x^3 - 2x - x$$

$$= 4x^3 - 3x$$

for 'N' odd

Let $N=1$

$$T_1(x) = x$$

for $x = \Omega = 0$

$$T_1(0) = 0$$

$$|H(0)|^2 = \frac{1}{1 + \epsilon^2 T_1^2(0)}$$

$$= \frac{1}{1 + \epsilon^2 \cdot 0}$$



$$|H(0)|^2 = 1$$

At $\Omega = 0$ gain is

max 1 and attenuation is zero

for 'N' even

Let $N=0$

$$T_0(x) = 1$$

Let $\Omega = x = 0$

$$T_0(0) = 1$$

$$|H(0)|^2 = \frac{1}{1 + \epsilon^2 T_0^2(0)}$$

$$= \frac{1}{1 + \epsilon^2 \cdot 1}$$

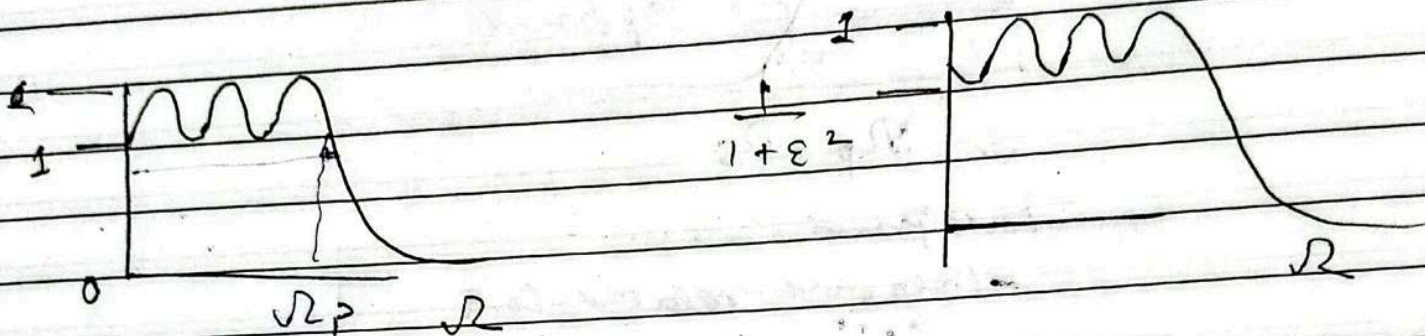
$$= \frac{1}{1 + \epsilon^2}$$

At $\Omega = 0$ gain is slightly

less than 1

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Res tha 1 depending on ϵ



At $\omega = \omega_p$

for odd

$$|H(j\omega)|^2 = \frac{1}{1 + \epsilon^2 \omega^2 \left(\frac{\omega_p}{\omega_p} \right)^2}$$

$$= \frac{1}{1 + \epsilon^2 \omega^2 (1)}$$

$$|H(j\omega)|^2 = \frac{1}{1 + \epsilon^2}$$

At $\omega = \omega_p$

for even

$$|H(j\omega)|^2 = \frac{1}{1 + \epsilon^2 \omega^2 (0)}$$

$$= \frac{1}{1 + \epsilon^2}$$

Hence we can write

$$|H(j\omega)| = \frac{1}{\sqrt{1 + \epsilon^2}} \approx (1 - \delta)$$

this make sum
then 1
then 2
this value
dp over δ

This is written
bck $\frac{1}{1+\epsilon}$

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$$\left(\frac{1}{\sqrt{1+\epsilon^2}} \right)^2 = (1-\delta_1)^2$$

$$\frac{1}{1+\epsilon^2} = (1-\delta_1)^2$$

$$\left[\epsilon^2 = \frac{1}{(1-\delta_1)^2} - 1 \right] \rightarrow (8.3.54)$$

δ_1 = value of the pass band ripple

Q. Design a FIR filter, approximating the ideal frequency response

$$H_d(\omega) = \begin{cases} 1 & , \quad |\omega| \leq \pi/6 \\ 0 & , \quad \pi/6 \leq \omega \leq \pi \end{cases}$$

Determine the coefficients of 25-tap filter by using hamming window.

~~What is the result~~ $h_d(n) = h(n) \cdot w(n)$

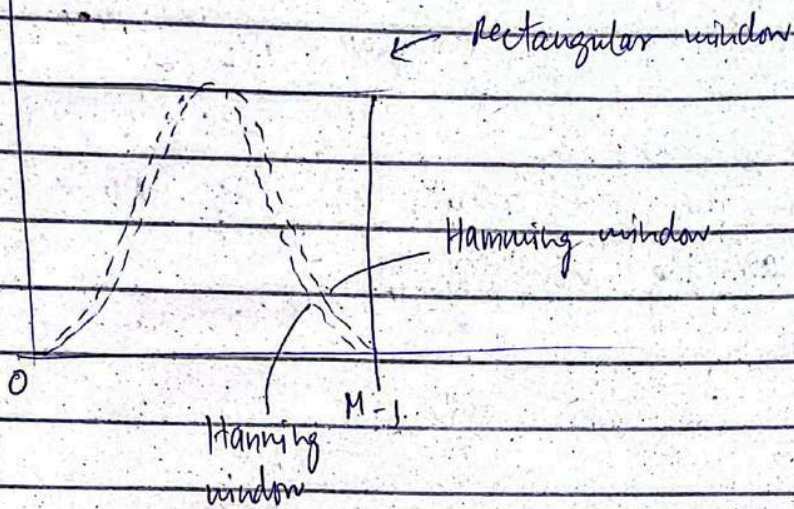
$$\begin{aligned} h(n) &= \frac{1}{2\pi} \int_{-\pi/6}^{\pi/6} H_d(\omega) e^{j\omega n} d\omega \times e^{-j\omega \frac{(25-1)}{2}} \\ &= \frac{1}{2\pi} \int_{-\pi/6}^{\pi/6} e^{j\omega n} d\omega = \frac{1}{2\pi} \left[\frac{e^{j\omega n}}{jn} \right]_{-\pi/6}^{\pi/6} \\ &= \frac{1}{2\pi} \int_{-\pi/6}^{\pi/6} e^{j\omega n} e^{-j\omega \frac{(25-1)}{2}} d\omega \\ &= \frac{1}{2\pi} \int_{-\pi/6}^{\pi/6} e^{j\omega (n-12)} d\omega \\ &= \frac{1}{2\pi} \left[\frac{e^{j\omega (n-12)}}{j(n-12)} \right]_{-\pi/6}^{\pi/6} \\ &= \frac{1}{2\pi} \frac{e^{j\pi/6 (n-12)} - e^{-j\pi/6 (n-12)}}{j(n-12)} \end{aligned}$$

Multiply & divide by 2

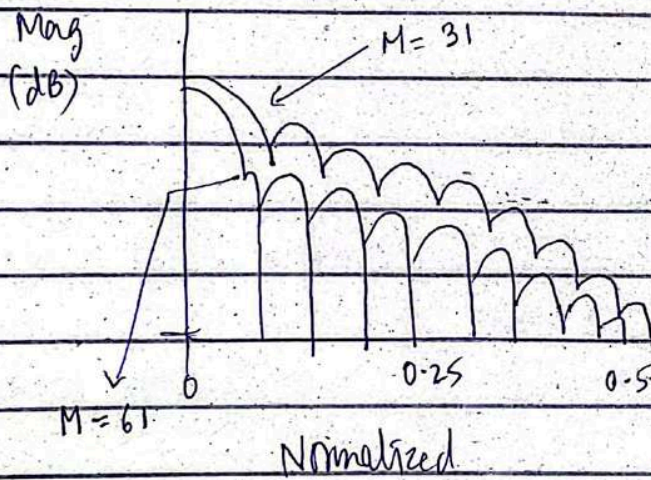
$$\begin{aligned} h(n) &= \frac{2}{2\pi (n-12)} \left[\frac{e^{j\pi/6 (n-12)} - e^{-j\pi/6 (n-12)}}{2j} \right] \\ &= \frac{\sin(\pi/6 (n-12))}{\pi (n-12)} \end{aligned}$$

Shapes of different windows:

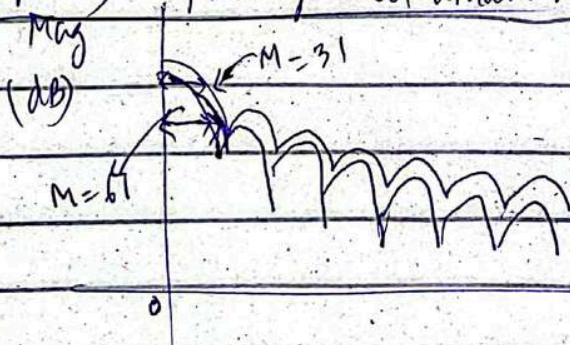
Magnitude



Frequency Response of Hanning window

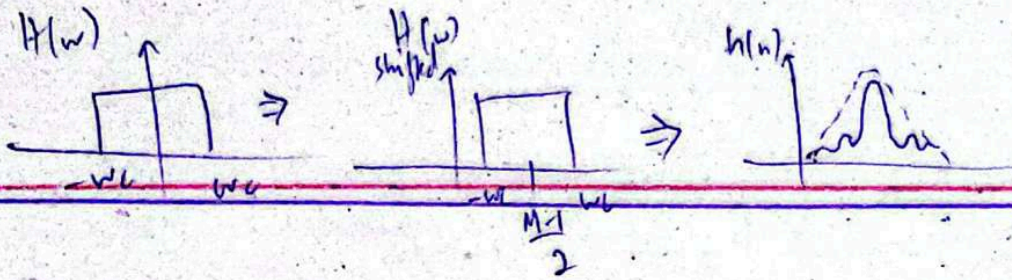


Frequency Response of Rect window:



→ sinc.
As we increase M ,
width increases.

- smaller values of M for rect & Hanning, response is same almost.
for higher values (61) it changes



$$h(n) = \frac{\sin \pi/6 (n-12)}{\pi(n-12)}$$

desired impulse response.

$$h_d(n) = h(n) \cdot w(n)$$

→ hamming window

→ $h(n)$ goes upto infinity, we need to truncate it so, we use windowing.

$$h_d(n) = \frac{\sin \pi/6 (n-12)}{\pi(n-12)} \cdot \left[0.54 - 0.46 \left(\frac{\cos(2\pi n)}{24} \right) \right]$$

↓
0 to 24.

$$w(n)_{\text{hamming}} = 0.54 - 0.46 \left(\frac{\cos 2\pi n}{24} \right)$$

$$w(n)_{\text{hanning}} = \frac{1}{2} \left(1 - \cos \frac{2\pi n}{M-1} \right)$$

for $h_d(12) \Rightarrow$ max. magnitude

